QP CODE: 24900153

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Multi-Disciplinary Course - MG1MDCJMC100 - BASICS OF PHOTOGRAPHY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest(I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability

[Learning Domain][CO No(s)]

Part A

Multiple Choice Questions Answer all questions.

Each question carries 1 mark

1. **b)** To guide the viewer's eye and create a message [U] [1]
2. **b)** Lighter weight and more compact [U] [1]
3. **b)** Healing brush [K] [1]
4. **b)** Instant photo development [U] [3]
5. **c)** Reflex mirror and optical viewfinder [U] [3]
6. **b)** It reduces shadows and creates a natural, gentle look [K] [4]
7. **b)** Exposure time [U] [3]
8. **c)** To bounce light onto the subject and fill in shadows [U] [3]
9. **b)** Just after sunrise and before sunset [U] [3]
10. **a)** To project patterns or shapes onto a surface [U] [4]
11. **a)** International Organization for Standardization [U] [4]
12. **c)** Create strong contrast [U] [4]
13. **b)** Manipulating and enhancing photographs or digital images [U] [5]
14. **d)** GIMP [U] [5]
15. **b)** Creating social media graphics [U] [5]

(1*15=15)

Part B

Short Answer Type Questions Answer any five questions.

Each question carries 3 marks

[U] [1]

16. Film cameras relied on chemical processes for image capture, while digital cameras use sensors that allow immediate image review and storage. The evolution of camera types from film to digital is as follows:

    - Early Film Cameras: Used photographic plates and later roll film. The Kodak box camera is one such example.
    - Compact 35mm Film Cameras: cameras with improved focusing and portability.
    - Polaroid Cameras (1948): Introduced instant photo development.
    - SLR Cameras (1940s): Featured single-lens reflex mechanisms, interchangeable lenses, and viewfinders.
    - DSLR Cameras (Mid-1990s): Combined digital sensors with SLR mechanisms, gaining widespread popularity.
    - Mirrorless Cameras (2010s): Eliminated the mirror mechanism, reducing size and weight while retaining high-quality performance.

[U] [1]

17. Limitations of Traditional Polaroid Cameras in Terms of Image Quality are as follows:
    - Lower image resolution compared to modern digital cameras or even traditional film cameras, resulting in less detailed images.
    - Polaroid film has a narrow dynamic range.
    - Polaroid cameras often have limited manual settings for exposure, leading to inconsistent results in varying lighting conditions.
    - softer images and reduced clarity due to the less advanced optics in traditional polaroid camera.
    - The images can appear grainy due to the chemical development process.
    - Colors in Polaroid images may not be as vibrant or accurate, sometimes appearing faded or oversaturated due to the chemical development process.

[U] [2]

18. Roles and Responsibilities of a photographer are as follows:
    - Capture High-Quality Images: Take well-composed and technically sound photos based on the client's or project's needs.
    - Manage Equipment: Select and use appropriate cameras, lenses, and lighting to achieve the desired effect.
    - Editing and Post-Processing: Enhance images through software to improve quality and meet specifications.
    - Ensure Ethical Practices: Respect privacy and seek consent, especially in portrait or sensitive photography.

- Maintain Client Communication: Understand client requirements and deliver photos within the set deadlines.

[U] [4]

19. Applying Symmetry to Achieve Balance in a Photograph:
    - Position subjects or objects on both sides of the frame to create a mirrored effect, ensuring visual harmony.
    - Place the main subject in the centre, with symmetrical elements radiating outward, giving a balanced and cohesive feel.
    - Align the subjects along a central vertical or horizontal axis to achieve balance on both sides of the frame.

[U] [3]

20. Key differences between SLRs and DSLRs:

    - SLRs uses film to capture images while DSLRs uses a digital sensor to capture images.
    - SLRs are Generally larger and heavier due to the film mechanism while DSLRs are Often bulkier but lighter than traditional film SLRs due to digital components.
    - No instant image review, requires developing the film in the case of SLRs. Instant image review on the LCD screen after each shot in DSLRs.
    - SLRs storage is limited to film roll length; no option for deleting or editing photos. Digital storage in DSLR allows unlimited shots and easy editing or deletion.

[U] [3]

21. Role of Shutter, Lens, and Aperture in a Camera:

    Shutter speed: It is the length of time the camera's shutter stays open, and therefore how long the sensor is exposed to light. The longer it's open, the more light hits the sensor and the brighter the image. A fast shutter speed, like 1/1000 of a second, can freeze the motion of a fast-moving subject. A slow shutter speed, like 1/30 of a second, can create motion blur.

    Lens: Focuses light onto the camera sensor, determining how much of the scene is visible in the frame and how sharp the subject appears.

    Aperture: The aperture of a camera lens is a hole that regulates how much light passes through to the film or sensor. Smaller apertures, like f/16, let in less light. Larger apertures like f/1.4 let in more light. It also affects the depth of field, controlling how much of the image is in focus. A larger aperture (1.4, 2 etc.) creates a shallow depth of field, while a smaller aperture (11,16) creates a greater depth of field.

[U] [4]

22. The exposure triangle consists of three variables that adjust how a camera captures light: aperture, shutter speed, and ISO. Together, these three elements properly expose a shot. The three variables of the exposure triangle are all dependent on each other.

Aperture: The aperture of a camera lens is a hole that regulates how much light passes through to the film or sensor. Smaller apertures, like f/16, let in less light. Larger apertures like f/1.4 let in more light. It also affects the depth of field, controlling how much of the image is in focus. A larger aperture (1.4, 2 etc.) creates a shallow depth of field, while a smaller aperture (11,16) creates a greater depth of field.

Shutter speed: It is the length of time the camera's shutter stays open, and therefore how long the sensor is exposed to light. The longer it's open, the more light hits the sensor and the brighter the image. A fast shutter speed, like 1/1000 of a second, can freeze the motion of a fast-moving subject. A slow shutter speed, like 1/30 of a second, can create motion blur.

ISO: A measure of how sensitive a camera's sensor is to light. ISO is measured in ISO values such as 50, 100, 200, 400, 800, 1600, 3200, etc. A lower ISO value such as 50 makes a camera's sensor *less* sensitive to light. A higher ISO value such as 1600 makes a camera's sensor *more* sensitive to light. Higher ISO values can add noise to an image. Increasing the ISO results in a brighter image without changing the shutter speed or aperture. Higher ISO is particularly useful in low light conditions.

Together, these settings determine the exposure (brightness), depth of field, and motion effects in the final image. Balancing them is crucial for achieving the desired look.

[U] [3]

23. Butterfly lighting is a popular lighting technique, especially in portrait photography. It involves positioning the light source above and slightly in front of the subject, creating a distinct shadow beneath the nose that resembles a butterfly shape. Key uses includes highlighting facial features, creating dramatic effects and are effective for close-up portraits, focusing on the subject's face and emphasizing features like the eyes, lips, and cheekbones.

(5 × 3 = 15)

Part C

Essay Type Questions
Answer any two questions.
Each question carries 10 marks

[A] [4]

24. Positive Space: Refers to the area of the image that contains the subject(s) or focal point(s). It is the main element that the viewer's attention is drawn to. Positive space gives the image substance, balance, and meaning. It is essential for conveying the subject's story or message. It focuses attention on what's important and helps define the purpose of the image.

Negative Space: Refers to the areas surrounding and between the subjects in the photograph. It often remains unoccupied or has minimal content. Negative space

helps to define and accentuate the positive space, creating a sense of openness, simplicity, and harmony. It creates breathing room around the subject, making the composition feel more balanced and less cluttered. It can also evoke emotions, such as calmness or isolation, and guide the viewer's gaze through the image.

Together, positive and negative spaces work to establish a visual balance, contrast, and flow in the image, ensuring the viewer's attention is directed in a deliberate way.

[An] [1]

25. Types of photography:

1. Portrait Photography
Portrait photography focuses on capturing the personality and mood of a subject, typically emphasizing the face.

2. Landscape Photography
Landscape photography captures natural scenery, including mountains, forests, rivers, deserts, and other outdoor environments. The aim is to convey the beauty and vastness of nature

3. Wildlife Photography
Wildlife photography focuses on capturing animals in their natural habitats. It requires patience and a deep understanding of animal behavior to capture dynamic, candid moments.

4. Documentary Photography
Documentary photography tells stories or provides information about an event, issue, or social situation. It often emphasizes realism and truth, with the aim to inform, raise awareness, or inspire change

5. Fashion Photography
Fashion photography is primarily focused on clothing, accessories, and beauty. The objective is to showcase fashion in a visually appealing way, often for advertisements, magazines, or catalogs

6. Still Life Photography
Still life photography involves photographing inanimate objects, often arranged in a controlled environment

7. Event Photography
Event photography captures moments from events such as weddings, parties, concerts, and corporate functions. The photographer's role is to document the atmosphere, people, and key moments of the event, often with a focus on candid shots as well as posed group photos.

8. Sports Photography
Sports photography focuses on capturing the action and energy of sporting events

9. Macro Photography
Macro photography involves taking close-up photographs of small subjects, such as insects, flowers, or textures.

10. Aerial Photography
Aerial photography is taken from an elevated perspective, typically using drones, airplanes, or helicopters.

[U] [3]

26. A tripod, filter and an external flash are essential accessories that play a vital role in enhancing the photographer's ability to produce high-quality, creative images.

- A tripod is a three-legged support used to stabilize the camera during shooting. It plays a crucial role in ensuring sharp, clear images by minimizing camera shake, especially in low-light conditions or when using slow shutter speeds. It helps with composition by allowing precise framing without the need for constant adjustments. It helps to enable creative techniques like time-lapse, panoramic, and astrophotography.
- Filters are used to modify the light entering the camera, enhancing specific aspects of the image or achieving special effects. There are different types of filters, such as polarizing filters, neutral density (ND) filters, and color filters. It controls light exposure and improves image quality by reducing glare and reflections. Provides flexibility in various lighting conditions, such as in bright sunlight or when shooting in harsh contrast.
- External flashes provide additional light when the available light is insufficient. Unlike built-in camera flashes, external flashes can be positioned and modified for more control over lighting. These flashes can be bounced off surfaces or diffused through modifiers like softboxes to create soft, flattering light.

[U] [3]

27. Special effects and filters play a multifaceted role in creative image editing, from enhancing image quality to transforming reality. By carefully choosing and applying these tools, photographers and editors can elevate their work, making it more engaging, expressive, and visually compelling.

Special effects like motion blur simulate movement, adding dynamism to action shots.

Filters such as sepia tones create a nostalgic or vintage atmosphere in portraits.

Vignette effects darken edges to draw focus to the central subject in product photography.

Color grading, such as cool blue tones, conveys calmness or melancholy in landscape images.

Grain filters mimic the texture of analog film for a retro aesthetic in fine art photography.

Brightness and contrast adjustments make underexposed images appear vibrant and balanced.

Bokeh overlays add creative light patterns, enhancing the artistic appeal of night photography.

Uniform filters ensure cohesive color palettes across social media posts, enhancing brand identity.

Textures like grunge overlays transform plain backgrounds into stylized, artistic elements.

Dramatic lighting effects, such as spotlight filters, add intensity to conceptual photography.

$(2 \times 10 = 20)$

***